

Practical Tools for Study Design and Inference

Chapter 6: Outcome Rates

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Use of Outcome Rates

- Outcome rates, such as the percent of sample units refusing to participate in a survey, generally have three uses.
 - ① Measure study performance. e.g., what proportion of the sample that resulted in a completed interview.
 - ② Inflate a calculated sample size for loss of sample units. For example, a survey statistician determines the number of sample units needed to detect a three percentage point difference in the estimates for males and females for a specified size and power of a test.
 - ③ Can be incorporated into the design weights as adjustment factors to create final analysis weights.

Disposition Codes

- Numeric codes that describe current or final data collection status of each sample unit
- AAPOR has a list of recommended disposition codes; however, codes are generally specific to each data collection agency.
- AAPOR Standard Definition: <https://aapor.org/wp-content/uploads/2024/03/Standards-Definitions-10th-edition.pdf>

Disposition Codes: CDC's BRFSS - 1

2020 Behavioral Risk Factor Surveillance System (BRFSS) Outcome Rates:

https://www.cdc.gov/brfss/annual_data/2020/pdf/2020-sdqr-508.pdf

Table 2. 2020 Disposition Codes for Landline Telephones and Cellular Telephones		
Category	Code	Description
Interviewed (1000-level codes)	1100	Completed interview
	1200	Partially completed interview
Eligible, Non-Interview (2000 level codes)	2111	Household level refusal (used for landline only)
	2112	Selected respondent refusal
	2120	Break off/termination within questionnaire
	2210	Selected respondent never available
	2320	Selected respondent physically or mentally unable to complete interview
	2330	Language barrier of selected respondent

Disposition Codes: CDC's BRFSS - 2

Behavioral Risk Factor Surveillance System (BRFSS) Outcome Rates:
https://www.cdc.gov/brfss/annual_data/2020/pdf/2020-sdqr-508.pdf

Table 2. 2020 Disposition Codes for Landline Telephones and Cellular Telephones		
Category	Code	Description
Unknown Eligibility	3100	Unknown if housing unit
	3130	No answer
	3140	Answering device, unknown whether eligible
	3150	Telecommunication barrier (i.e. call blocking)
	3200	Household, not known if respondent eligible
	3322	Physical or mental impairment (household level)
	3330	Language barrier (household level)
	3700	On never-call list
Not Eligible	4100	Out of sample
	4200	Fax/data/modem
	4300	Nonworking/disconnected number
	4400	Technological barrier (i.e., fast busy, phone circuit barriers)
	4430	Call forwarding/pager
	4460	Landline telephone number (used for cellular telephone only)
	4500	Non-residence/business
	4900	Miscellaneous, non-eligible

Study Rates

There are several rates defined by AAPOR.

- Location
- Contact
- Eligibility
- Cooperation
- Response

Formulas in VDK Chapter 6.2

- $RR1 = \frac{I}{I + P + R + NC + O + U}$
- $RR2 = \frac{I + P}{I + P + R + NC + O + U}$

Study Rates & Eligibility

Table 6.2: Disposition code categories used in performance rate calculation

Symbol in study rates	Category	Study eligibility
<i>I</i>	Complete interview	<i>Eligible</i>
<i>P</i>	Partially complete interview	<i>Eligible</i>
<i>R</i>	Refusal/break-off	<i>Eligible</i>
<i>NC</i>	Noncontact	<i>Eligible</i>
<i>U</i>	Unknown study eligibility	<i>Unknown</i>
<i>NE</i>	Not eligible	<i>Ineligible</i>
<i>O</i>	Other	<i>Eligible</i>

Use Rates to Adjust Initial Sample Sizes

- Decide on desired final sample size of respondents
- Make your best guesstimates about loss rates at different stages of data collection
Having best-case and worst-case scenarios is advisable
- Set up a spreadsheet that will calculate the required initial sample size and show how much the sample is reduced for each kind of loss

Inflating Sample Sizes for Losses

Table 6.6: Example of inflating target interviews for sample loss

Target interviews	200
<i>Cooperation rate</i>	<i>0.7–0.75</i>
Number of eligibles	267–286
<i>Eligibility rate</i>	<i>0.8–0.85</i>
Number screened	314–357
<i>Screening rate</i>	<i>0.7–0.82</i>
Number contacted	383–510
<i>Contact rate</i>	<i>0.92–0.95</i>
Occupied HUs	403–555
<i>Occupied HU rate</i>	<i>0.95–0.97</i>
Sample addresses	415–584

Set up a spreadsheet with anticipated outcome rates. Plug in Target Interviews for automatic computation of initial sample size.

Replicates - 1

- Randomly subdivide the full sample into data collection subsamples (under a "worst case scenario") called **replicates**
- Release only the number of replicates necessary to meet the study goals
- Easy example
 - *srswor* of 500 members of a professional association is selected with the goal being to obtain 100 completed questionnaires (completes).
 - Divide the 500 randomly into 10 replicates of size 50.
 - Initially, release first three replicates
 - If necessary, additional replicates are released to obtain the desired 100 completes.
- Replicates do not need to contain same number of cases. However, having equal-sized replicates is convenient and makes accounting easier.

Replicates - 2

- If replicates are randomly constructed, deciding to withhold any replicate does not negate the randomness of the sample; weights are adjusted to reflect only released replicates.
- However, once a replicate has been released, all cases in that replicate must be worked and be given a disposition code. Alternatively, the cases could be worked in a random order. Here, data collection could be stopped partway through a replicate, but this is impractical.
- Creating replicates in advance is much easier than selecting an initial sample and then attempting to add to it later, depending on the sample design.
 - Adding to a simple random sample by selecting another *srs* from the initial non-sample units is legitimate.
 - But, if the initial sample is selected with probabilities proportional to size (*pps*), selecting a supplemental sample in such a way that the overall sample is *pps* is not straightforward.

Rules of Thumb for Costs

- Having some rules-of-thumb for time required for different tasks is useful when estimating costs
- Estimate hours for tasks then convert to \$
- Telephone survey example
 - Interviewers spend time on non-interview tasks (training, filling out timesheets, unproductive calls, etc.)
 - Total interviewer hours = (no. of intvs) $\times$ (time to do intv) $\times$ (factor to account for administrative time)
 - No. of staff needed = (total no. of intv hours) / (weeks in data collection period) / (average no. of hours intvrs work per week)
 - Supervisory hours = 15 or 20% of interviewer hours
 - Management and facilities costs
 - Overhead (indirect) costs: cost of staff benefits; organizations usually use a percentage of direct costs